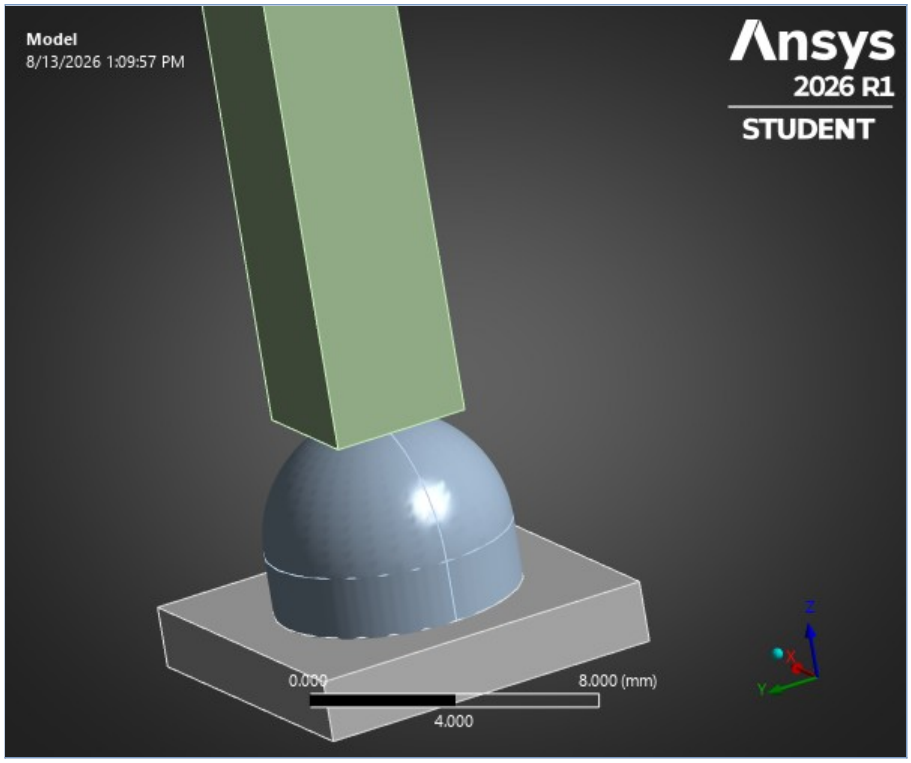
# Project*

| | |
|---|---|
| First Saved | Tuesday, August 11, 2026 |
| Last Saved | Tuesday, August 11, 2026 |
| Product Version | 2026 R1 |
| Save Project Before Solution | No |
| Save Project After Solution | No |



# Contents


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      - Results
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        - Status
      - Probes
- **Material Data**
  - ecoflex 00-45
  - Plastic, ABS (high-impact)


# Units

**TABLE 1**

| | |
|---|---|
| Unit System | Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius |
| Angle | Degrees |
| Rotational Velocity | rad/s |
| Temperature | Celsius |

# Model (A4)

**TABLE 2**
**Model (A4) > Geometry Imports**

| | |
|---|---|
| Object Name | *Geometry Imports* |
| State | Solved |

**TABLE 3**
**Model (A4) > Geometry Imports > Geometry Import (A3)**

| | |
|---|---|
| Object Name | *Geometry Import (A3)* |
| State | Solved |
| **Definition** | |
| Source | C:\Users\DLSU\Downloads\run 6 solid.step |
| Type | Step |
| **Basic Geometry Options** | |
| Solid Bodies | Yes |
| Surface Bodies | Yes |
| Line Bodies | No |
| Parameters | Independent |
| Parameter Key | ANS;DS |
| Attributes | No |
| Named Selections | No |
| Material Properties | No |
| **Advanced Geometry Options** | |
| Use Associativity | Yes |
| Coordinate Systems | No |
| Reader Mode Saves Updated File | No |
| Use Instances | Yes |
| Smart CAD Update | Yes |
| Compare Parts On Update | No |

| | |
|---|---|
| Analysis Type | 3-D |
| Mixed Import Resolution | None |
| Import Facet Quality | Source |
| Clean Bodies On Import | No |
| Stitch Surfaces On Import | None |
| Decompose Disjoint Geometry | Yes |
| Enclosure and Symmetry Processing | Yes |

## *Geometry*

**TABLE 4**
**Model (A4) > Geometry**

| | |
|---|---|
| Object Name | *Geometry* |
| State | Fully Defined |
| **Definition** | |
| Source | C:\Users\DLSU\Downloads\run 6 solid.step |
| Type | Step |
| Length Unit | Millimeters |
| Element Control | Program Controlled |
| Display Style | Body Color |
| **Bounding Box** | |
| Length X | 10. mm |
| Length Y | 10. mm |
| Length Z | 45.05 mm |
| **Properties** | |
| Volume | 940.03 mm³ |
| Mass | 9.7819e-004 kg |
| Scale Factor Value | 1. |
| **Statistics** | |
| Bodies | 3 |
| Active Bodies | 3 |
| Nodes | 1916 |
| Elements | 520 |
| Mesh Metric | None |
| **Update Options** | |
| Assign Default Material | No |
| **Basic Geometry Options** | |
| Solid Bodies | Yes |
| Surface Bodies | Yes |
| Line Bodies | No |
| Parameters | Independent |
| Parameter Key | ANS;DS |
| Attributes | No |
| Named Selections | No |
| Material Properties | No |
| **Advanced Geometry Options** | |
| Use Associativity | Yes |
| Coordinate Systems | No |
| Reader Mode Saves Updated File | No |
| Use Instances | Yes |
| Smart CAD Update | Yes |
| Compare Parts On Update | No |
| Analysis Type | 3-D |
| Mixed Import Resolution | None |
| Import Facet Quality | Source |
| Clean Bodies On Import | No |
| Stitch Surfaces On Import | None |
| Decompose Disjoint Geometry | Yes |
| ID_GeometryPrefProcessPhysicsDefinition | No |
| Enclosure and Symmetry Processing | Yes |

**TABLE 5**
**Model (A4) > Geometry > Parts**

| Object Name | *run 6 solid\|Body1* | *run 6 solid\|Body3* | *run 6 solid\|Body2* |
|---|---|---|---|
| State | Meshed | | |
| **Graphics Properties** | | | |
| Visible | Yes | | |
| Transparency | 1 | | |
| **Definition** | | | |
| Suppressed | No | | |

| | | | |
|---|---|---|---|
| Stiffness Behavior | Flexible | | |
| Coordinate System | Default Coordinate System | | |
| Reference Temperature | By Environment | | |
| Treatment | None | | |
| **Material** | | | |
| Assignment | ecoflex 00-45 | | Plastic, ABS (high-impact) |
| Nonlinear Effects | Yes | | |
| Thermal Strain Effects | Yes | | |
| **Bounding Box** | | | |
| Length X | 10. mm | 7. mm | 4. mm |
| Length Y | 10. mm | 7. mm | 4. mm |
| Length Z | 1.75 mm | 7. mm | 38. mm |
| **Properties** | | | |
| Volume | 107.68 mm³ | 224.35 mm³ | 608. mm³ |
| Mass | 1.1414e-004 kg | 2.3781e-004 kg | 6.2624e-004 kg |
| Centroid X | -1.2364e-016 mm | 1.3018e-011 mm | 0. mm |
| Centroid Y | -1.5466e-017 mm | 5.385e-011 mm | 7.791e-018 mm |
| Centroid Z | 0.875 mm | 2.9745 mm | 26.05 mm |
| Moment of Inertia Ip1 | 1.3565e-003 kg·mm² | 1.4197e-003 kg·mm² | 7.6193e-002 kg·mm² |
| Moment of Inertia Ip2 | 1.3565e-003 kg·mm² | 1.4197e-003 kg·mm² | 7.6193e-002 kg·mm² |
| Moment of Inertia Ip3 | 2.6548e-003 kg·mm² | 1.3394e-003 kg·mm² | 1.67e-003 kg·mm² |
| **Statistics** | | | |
| Nodes | 632 | 723 | 561 |
| Elements | 80 | 368 | 72 |
| Mesh Metric | None | | |

**TABLE 6**
**Model (A4) > Materials**

| | |
|---|---|
| Object Name | *Materials* |
| State | Fully Defined |
| **Statistics** | |
| Materials | 3 |
| Material Assignments | 0 |

## *Coordinate Systems*

**TABLE 7**
**Model (A4) > Coordinate Systems > Coordinate System**

| | |
|---|---|
| Object Name | *Global Coordinate System* |
| State | Fully Defined |
| **Definition** | |
| Type | Cartesian |
| Coordinate System ID | 0. |
| **Origin** | |
| Origin X | 0. mm |
| Origin Y | 0. mm |
| Origin Z | 0. mm |
| **Directional Vectors** | |
| X Axis Data | [ 1. 0. 0. ] |
| Y Axis Data | [ 0. 1. 0. ] |
| Z Axis Data | [ 0. 0. 1. ] |
| **Transfer Properties** | |
| Source | |
| Read Only | No |

## *Connections*

**TABLE 8**
**Model (A4) > Connections**

| | |
|---|---|
| Object Name | *Connections* |
| State | Fully Defined |
| **Auto Detection** | |
| Generate Automatic Connection On Refresh | Yes |
| **Transparency** | |
| Enabled | Yes |
| **Statistics** | |
| Contacts | 3 |
| Active Contacts | 2 |
| Joints | 0 |
| Active Joints | 0 |

| | |
|---|---|
| Beams | 0 |
| Active Beams | 0 |
| Bearings | 0 |
| Active Bearings | 0 |
| Springs | 0 |
| Active Springs | 0 |
| Body Interactions | 0 |
| Active Body Interactions | 0 |

**TABLE 9**
**Model (A4) > Connections > Contacts**

| Object Name | *Contacts* | *Contacts 2* |
|---|---|---|
| State | Fully Defined | |
| **Definition** | | |
| Connection Type | Contact | |
| **Scope** | | |
| Scoping Method | Geometry Selection | |
| Geometry | All Bodies | |
| **Auto Detection** | | |
| Tolerance Type | Slider | |
| Tolerance Slider | 0. | |
| Tolerance Value | 0.11804 mm | |
| Use Range | No | |
| Face/Face | Yes | |
| Face-Face Angle Tolerance | 75. ° | |
| Face Overlap Tolerance | Off | |
| Cylindrical Faces | Include | |
| Face/Edge | No | |
| Edge/Edge | No | |
| Priority | Include All | |
| Group By | Bodies | |
| Search Across | Bodies | |
| **Statistics** | | |
| Connections | 2 | 1 |
| Active Connections | 1 | |

**TABLE 10**
**Model (A4) > Connections > Contacts > Contact Regions**

| Object Name | *Contact Region* | *Contact Region 2* |
|---|---|---|
| State | Fully Defined | Suppressed |
| **Scope** | | |
| Scoping Method | Geometry Selection | |
| Contact | 1 Face | |
| Target | 1 Face | |
| Contact Bodies | run 6 solid\|Body1 | run 6 solid\|Body3 |
| Target Bodies | run 6 solid\|Body3 | run 6 solid\|Body2 |
| Protected | No | |
| **Definition** | | |
| Type | Bonded | |
| Scope Mode | Automatic | |
| Behavior | Program Controlled | |
| Trim Contact | Program Controlled | |
| Trim Tolerance | 0.11804 mm | |
| Contact APDL Name | | |
| Target APDL Name | | |
| Suppressed | No | Yes |
| **Display** | | |
| Element Normals | No | |
| **Advanced** | | |
| Formulation | Program Controlled | |
| Small Sliding | Program Controlled | |
| Detection Method | Program Controlled | |
| Penetration Tolerance | Program Controlled | |
| Elastic Slip Tolerance | Program Controlled | |
| Normal Stiffness | Program Controlled | |
| Update Stiffness | Program Controlled | |
| Pinball Region | Program Controlled | |
| **Geometric Modification** | | |
| Contact Geometry Correction | None | |
| Target Geometry Correction | None | |

**TABLE 11**
**Model (A4) > Connections > Contacts 2 > Contact Regions**

| | |
|---|---|
| Object Name | *Frictionless - run 6 solid\|Body3 To run 6 solid\|Body2* |
| State | Fully Defined |
| **Scope** | |
| Scoping Method | Geometry Selection |
| Contact | 1 Face |
| Target | 1 Face |
| Contact Bodies | run 6 solid\|Body3 |
| Target Bodies | run 6 solid\|Body2 |
| Protected | No |
| **Definition** | |
| Type | Frictionless |
| Scope Mode | Manual |
| Behavior | Asymmetric |
| Trim Contact | Program Controlled |
| Contact APDL Name | |
| Target APDL Name | |
| Suppressed | No |
| **Display** | |
| Element Normals | No |
| **Advanced** | |
| Formulation | Augmented Lagrange |
| Small Sliding | Program Controlled |
| Detection Method | Program Controlled |
| Penetration Tolerance | Program Controlled |
| Normal Stiffness | Program Controlled |
| Update Stiffness | Program Controlled |
| Stabilization Damping Factor | 0. |
| Pinball Region | Program Controlled |
| Time Step Controls | None |
| **Geometric Modification** | |
| Interface Treatment | Add Offset, No Ramping |
| Offset | 0. mm |
| Contact Geometry Correction | None |
| Target Geometry Correction | None |

## *Mesh*

**TABLE 12**
**Model (A4) > Mesh**

| | |
|---|---|
| Object Name | *Mesh* |
| State | Solved |
| **Display** | |
| Display Style | Use Geometry Setting |
| **Defaults** | |
| Physics Preference | Mechanical |
| Element Order | Program Controlled |
| Element Size | Default |
| **Sizing** | |
| Mesh Sizing | Adaptive |
| Resolution | Default (2) |
| Mesh Defeaturing | Yes |
| Defeature Size | Default |
| Transition | Fast |
| Span Angle Center | Coarse |
| Initial Size Seed | Assembly |
| Bounding Box Diagonal | 47.218 mm |
| Average Surface Area | 66.494 mm² |
| Minimum Edge Length | 1.75 mm |
| **Quality** | |
| Check Mesh Quality | Yes, Errors |
| Error Limits | Aggressive Mechanical |
| Target Element Quality | Default (0.1) |
| Smoothing | Medium |
| Mesh Metric | None |
| **Inflation** | |
| Use Automatic Inflation | None |
| Inflation Option | Smooth Transition |
| Transition Ratio | 0.272 |
| | |

| | |
|---|---|
| Maximum Layers | 5 |
| Growth Rate | 1.2 |
| Inflation Algorithm | Pre |
| Inflation Element Type | Wedges |
| View Advanced Options | No |
| **Advanced** | |
| Number of CPUs for Parallel Part Meshing | Program Controlled |
| Straight Sided Elements | No |
| Rigid Body Behavior | Dimensionally Reduced |
| Triangle Surface Mesher | Program Controlled |
| Topology Checking | Yes |
| Pinch Tolerance | Please Define |
| Generate Pinch on Refresh | No |
| Auto-Map Fillets | No |
| **Automatic Methods** | |
| Sheet Body Method | Prime Quad Dominant |
| Sweepable Body Method | Sweep |
| **Statistics** | |
| Nodes | 1916 |
| Elements | 520 |
| Show Detailed Statistics | No |

## Static Structural (A5)

**TABLE 13**
**Model (A4) > Analysis**

| | |
|---|---|
| Object Name | *Static Structural (A5)* |
| State | Solved |
| **Definition** | |
| Physics Type | Structural |
| Analysis Type | Static Structural |
| Solver Target | Mechanical APDL |
| **Options** | |
| Environment Temperature | 22. °C |
| Generate Input Only | No |

**TABLE 14**
**Model (A4) > Static Structural (A5) > Analysis Settings**

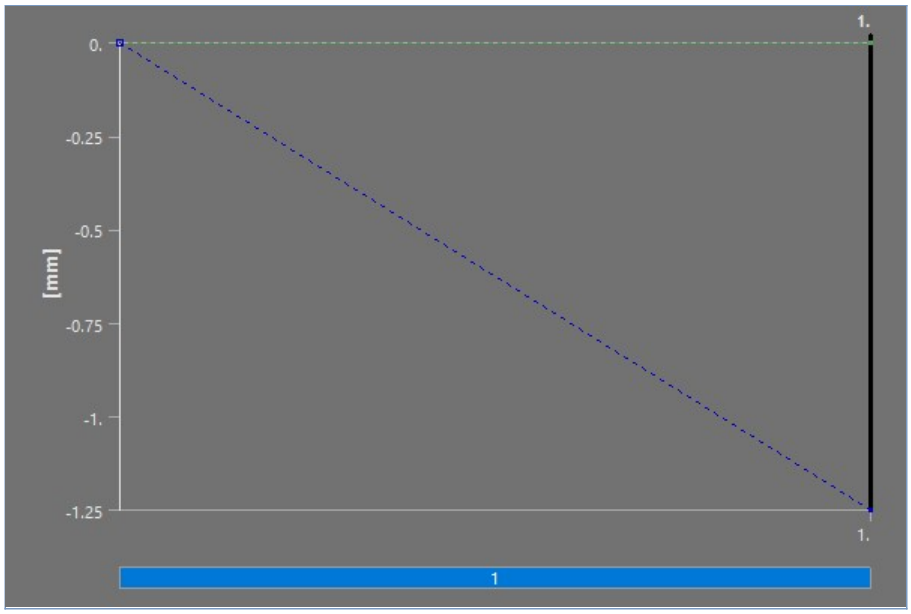
| | |
|---|---|
| Object Name | *Analysis Settings* |
| State | Fully Defined |
| **Step Controls** | |
| Number Of Steps | 1. |
| Current Step Number | 1. |
| Step End Time | 1. s |
| Auto Time Stepping | On |
| Define by | Substeps |
| Initial Substeps | 20. |
| Minimum Substeps | 5. |
| Maximum Substeps | 500. |
| Maximum Iterations Per Substep | Program Controlled |
| **Solver Controls** | |
| Solver Type | Program Controlled |
| Weak Springs | On |
| Spring Stiffness | Program Controlled |
| Solver Pivot Checking | Program Controlled |
| Large Deflection | On |
| Inertia Relief | Off |
| Quasi-Static Solution | Off |
| **Rotordynamics Controls** | |
| Coriolis Effect | Off |
| **Restart Controls** | |
| Generate Restart Points | Program Controlled |
| Retain Files After Full Solve | No |
| Combine Restart Files | Program Controlled |
| **Nonlinear Controls** | |
| Newton-Raphson Option | Program Controlled |
| Force Convergence | Program Controlled |
| Moment Convergence | Program Controlled |
| Displacement Convergence | Program Controlled |

| | |
|---|---|
| Rotation Convergence | Program Controlled |
| Line Search | Program Controlled |
| Stabilization | Program Controlled |
| **Advanced** | |
| Inverse Option | No |
| Contact Split (DMP) | Program Controlled |
| **Output Controls** | |
| Output Selection | None |
| Stress | Yes |
| Back Stress | No |
| Strain | Yes |
| Contact Data | Yes |
| Nonlinear Data | No |
| Nodal Forces | No |
| Volume and Energy | Yes |
| Euler Angles | Yes |
| General Miscellaneous | No |
| Contact Miscellaneous | No |
| Store Results At | All Time Points |
| **Result File Options** | |
| Result File Compression | Program Controlled |
| **Analysis Data Management** | |
| Solver Files Directory | C:\Users\DLSU\Downloads\THESIS DATA\ANSYS FOR HOLLOW RODS\solid v hollow and diameter_files\dp0 \SYS\MECH\ |
| Future Analysis | None |
| Scratch Solver Files Directory | |
| Save MAPDL db | No |
| Contact Summary | Program Controlled |
| Delete Unneeded Files | Yes |
| Nonlinear Solution | Yes |
| Solver Units | Active System |
| Solver Unit System | nmm |

**TABLE 15**
**Model (A4) > Static Structural (A5) > Loads**

| Object Name | *Fixed Support* | *Displacement* |
|---|---|---|
| State | Fully Defined | |
| **Scope** | | |
| Scoping Method | Geometry Selection | |
| Geometry | 2 Faces | 1 Face |
| **Definition** | | |
| Type | Fixed Support | Displacement |
| Suppressed | No | |
| Define by | | Components |
| Coordinate System | | Global Coordinate System |
| X Component | | 0. mm (ramped) |
| Y Component | | 0. mm (ramped) |
| Z Component | | -1.25 mm (ramped) |

**FIGURE 1**
**Model (A4) > Static Structural (A5) > Displacement**




## *Solution (A6)*

**TABLE 16**
**Model (A4) > Static Structural (A5) > Solution**

| Object Name | *Solution (A6)* |
|---|---|
| State | Solved |
| **Adaptive Mesh Refinement** | |
| Max Refinement Loops | 1. |
| Refinement Depth | 2. |
| **Information** | |
| Status | Done |
| MAPDL Elapsed Time | 22.817 s |
| MAPDL Memory Used | 197. MB |
| MAPDL Result File Size | 3.1875 MB |
| **Post Processing** | |
| Beam Section Results | No |
| On Demand Stress/Strain | No |

**TABLE 17**
**Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information**

| Object Name | *Solution Information* |
|---|---|
| State | Solved |
| **Solution Information** | |
| Solution Output | Solver Output |
| Newton-Raphson Residuals | 0 |
| Identify Element Violations | 0 |
| Update Interval | 2.5 s |
| Display Points | All |
| **FE Connection Visibility** | |
| Activate Visibility | Yes |
| Display | All FE Connectors |
| Draw Connections Attached To | All Nodes |
| Line Color | Connection Type |
| Visible on Results | No |
| Line Thickness | Single |
| Display Type | Lines |

**TABLE 18**
**Model (A4) > Static Structural (A5) > Solution (A6) > Results**

| Object Name | *Total Deformation* | *Directional Deformation* | *Directional Deformation 2* | *Directional Deformation 3* | *Equivalent Elastic Strain* | *Maximum Principal Elastic Strain* | *Maximum Shear Elastic Strain* | *Equivalent Stress* | *Maximum Principal Stress* | *Maximum Shear Stress* | *Stress Intensity* |
|---|---|---|---|---|---|---|---|---|---|---|---|
| State | Solved | | | | | | | | | | |
| **Scope** | | | | | | | | | | | |
| Scoping | Geometry Selection | | | | | | | | | | |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Method | | | | | | | | | | | |
| Geometry | 1 Body | | | | | | | | | | |
| **Definition** | | | | | | | | | | | |
| Type | Total Deformation | Directional Deformation | | | Equivalent Elastic Strain | Maximum Principal Elastic Strain | Maximum Shear Elastic Strain | Equivalent (von-Mises) Stress | Maximum Principal Stress | Maximum Shear Stress | Stress Intensity |
| By | Time | | | | | | | | | | |
| Display Time | Last | | | | | | | | | | |
| Separate Data by Entity | No | | | | | | | | | | |
| Calculate Time History | Yes | | | | | | | | | | |
| Identifier | | | | | | | | | | | |
| Suppressed | No | | | | | | | | | | |
| Orientation | | X Axis | Y Axis | Z Axis | | | | | | | |
| Coordinate System | | Global Coordinate System | | | | | | | | | |
| **Results** | | | | | | | | | | | |
| Minimum | 0. mm | -0.2485 mm | -0.24997 mm | -1.1627 mm | 2.8821e-002 mm/mm | 9.98e-003 mm/mm | 3.2066e-002 mm/mm | 2.982e-003 MPa | -5.4577e-002 MPa | 1.4946e-003 MPa | 2.9891e-003 MPa |
| Maximum | 1.1628 mm | 0.23449 mm | 0.23835 mm | 0. mm | 0.29203 mm/mm | 0.13808 mm/mm | 0.41654 mm/mm | 3.5227e-002 MPa | 8.4568e-003 MPa | 1.7711e-002 MPa | 3.5422e-002 MPa |
| Average | 0.30562 mm | 4.4207e-003 mm | -3.7667e-003 mm | -0.25089 mm | 0.10035 mm/mm | 5.9054e-002 mm/mm | 0.15148 mm/mm | 1.2817e-002 MPa | -1.482e-003 MPa | 6.9066e-003 MPa | 1.3813e-002 MPa |
| Minimum Occurs On | run 6 solid\|Body3 | | | | | | | | | | |
| Maximum Occurs On | run 6 solid\|Body3 | | | | | | | | | | |
| **Minimum Value Over Time** | | | | | | | | | | | |
| Minimum | 0. mm | -0.2485 mm | -0.24997 mm | -1.1627 mm | 3.4967e-016 mm/mm | -5.3291e-016 mm/mm | 6.9389e-018 mm/mm | 1.5321e-017 MPa | -5.4577e-002 MPa | 8.845e-018 MPa | 1.769e-017 MPa |
| Maximum | 0. mm | -1.8022e-015 mm | -2.5883e-015 mm | -2.0357e-015 mm | 2.8821e-002 mm/mm | 9.98e-003 mm/mm | 3.2066e-002 mm/mm | 2.982e-003 MPa | -9.3872e-015 MPa | 1.4946e-003 MPa | 2.9891e-003 MPa |
| **Maximum Value Over Time** | | | | | | | | | | | |
| Minimum | 7.706e-015 mm | 6.8605e-015 mm | 1.8793e-015 mm | 0. mm | 3.7822e-015 mm/mm | 2.5535e-015 mm/mm | 5.033e-015 mm/mm | 6.1801e-016 MPa | 7.3559e-015 MPa | 3.4715e-016 MPa | 6.943e-016 MPa |
| Maximum | 1.1628 mm | 0.23449 mm | 0.23835 mm | 4.2284e-015 mm | 0.29203 mm/mm | 0.13808 mm/mm | 0.41654 mm/mm | 3.5227e-002 MPa | 8.4568e-003 MPa | 1.7711e-002 MPa | 3.5422e-002 MPa |
| **Information** | | | | | | | | | | | |
| Time | 1. s | | | | | | | | | | |
| Load Step | 1 | | | | | | | | | | |
| Substep | 8 | | | | | | | | | | |
| Iteration Number | 29 | | | | | | | | | | |
| **Integration Point Results** | | | | | | | | | | | |
| Display Option | | | | | Averaged | | | | | | |
| Average Across Bodies | | | | | No | | | | | | |

**FIGURE 2**
**Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation**

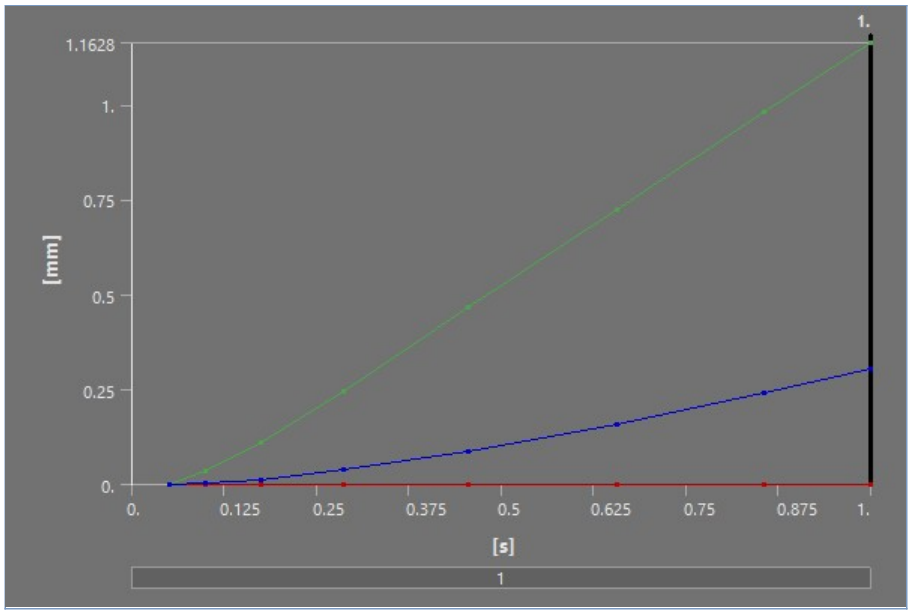



**TABLE 19**
**Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation**

| Time [s] | Minimum [mm] | Maximum [mm] | Average [mm] |
|---|---|---|---|
| 5.e-002 | 0. | 7.706e-015 | 1.2846e-015 |
| 0.1 | | 3.6776e-002 | 3.9722e-003 |
| 0.175 | | 0.10896 | 1.2505e-002 |
| 0.2875 | | 0.24551 | 3.7622e-002 |
| 0.45625 | | 0.46599 | 8.8236e-002 |
| 0.65625 | | 0.7237 | 0.15961 |
| 0.85625 | | 0.97982 | 0.24115 |
| 1. | | 1.1628 | 0.30562 |

**FIGURE 3**
**Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation > Figure**

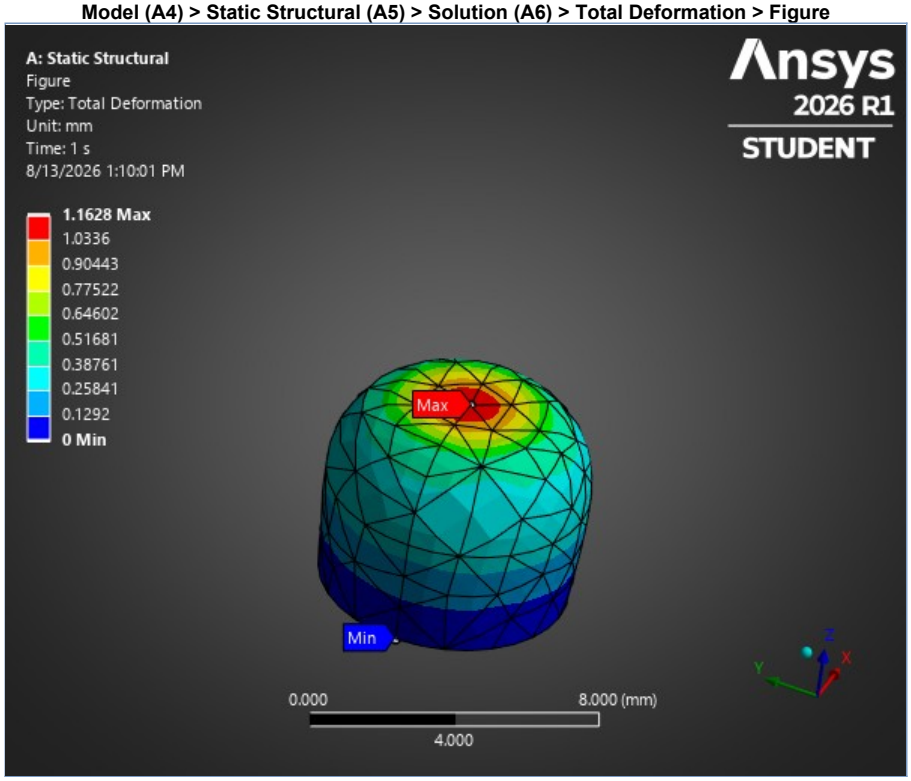



**FIGURE 4**
**Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation**

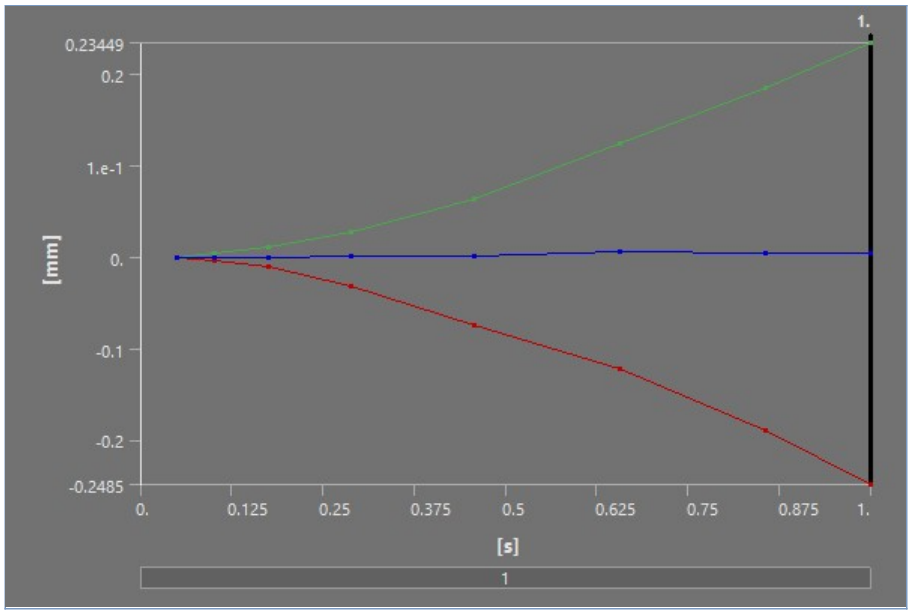



**TABLE 20**
**Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation**

| Time [s] | Minimum [mm] | Maximum [mm] | Average [mm] |
|---|---|---|---|
| 5.e-002 | -1.8022e-015 | 6.8605e-015 | 3.633e-016 |
| 0.1 | -3.2693e-003 | 3.7037e-003 | 1.6324e-004 |
| 0.175 | -1.0395e-002 | 1.0297e-002 | 3.0421e-004 |
| 0.2875 | -3.1086e-002 | 2.7382e-002 | 6.5733e-004 |
| 0.45625 | -7.4392e-002 | 6.3882e-002 | 4.6452e-004 |
| 0.65625 | -0.12202 | 0.1244 | 5.4532e-003 |
| 0.85625 | -0.19007 | 0.18601 | 5.2249e-003 |
| 1. | -0.2485 | 0.23449 | 4.4207e-003 |

**FIGURE 5**
**Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation > Figure**

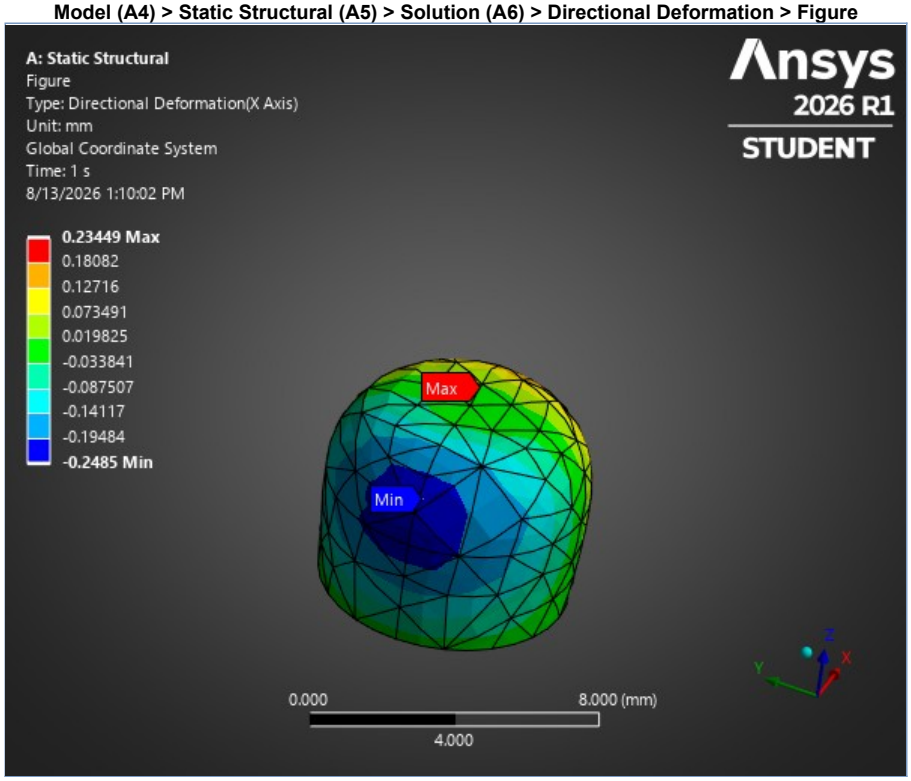



**FIGURE 6**
**Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation 2**

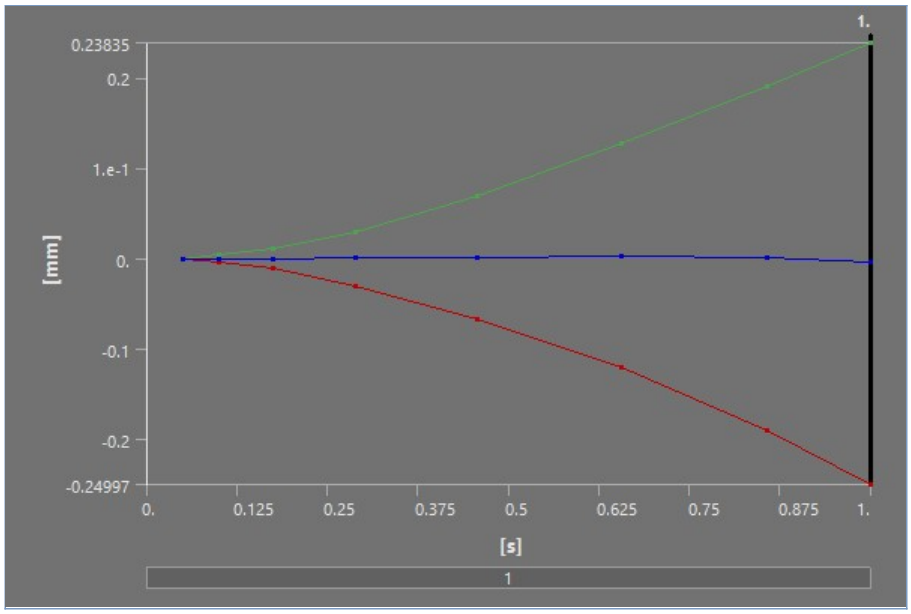



**TABLE 21**
**Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation 2**

| Time [s] | Minimum [mm] | Maximum [mm] | Average [mm] |
|---|---|---|---|
| 5.e-002 | -2.5883e-015 | 1.8793e-015 | -1.5508e-016 |
| 0.1 | -3.7327e-003 | 3.5813e-003 | -8.783e-005 |
| 0.175 | -1.1244e-002 | 1.1512e-002 | -1.316e-004 |
| 0.2875 | -2.9971e-002 | 2.9322e-002 | 3.8892e-004 |
| 0.45625 | -6.7795e-002 | 6.8782e-002 | 8.9552e-004 |
| 0.65625 | -0.12089 | 0.1267 | 2.7769e-003 |
| 0.85625 | -0.19002 | 0.19048 | 3.1531e-004 |
| 1. | -0.24997 | 0.23835 | -3.7667e-003 |

**FIGURE 7**
**Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation 2 > Figure**

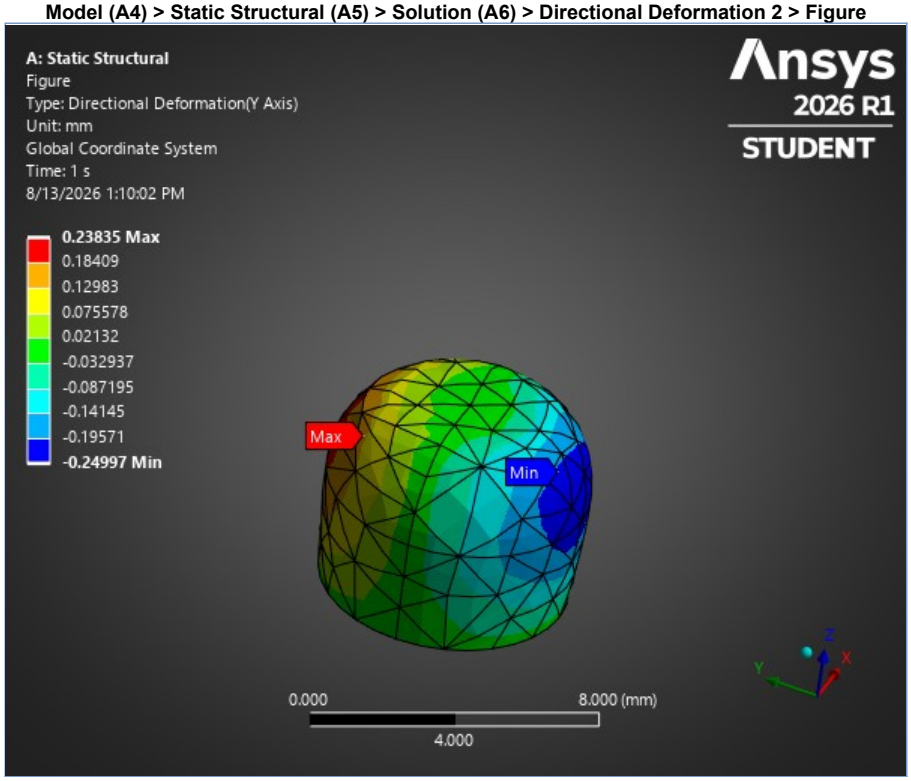



**FIGURE 8**
**Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation 3**

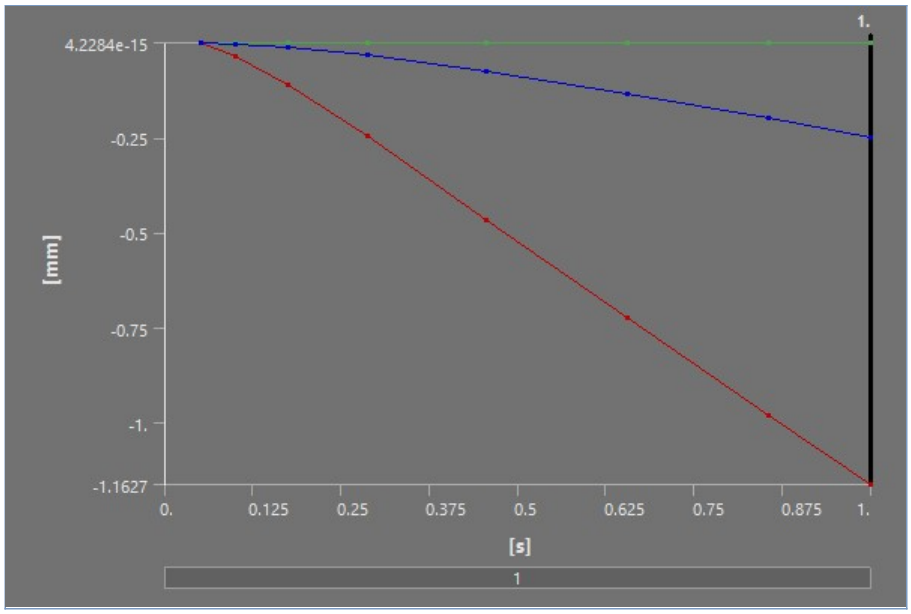



**TABLE 22**
**Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation 3**

| Time [s] | Minimum [mm] | Maximum [mm] | Average [mm] |
|---|---|---|---|
| 5.e-002 | -2.0357e-015 | 4.2284e-015 | 5.8156e-016 |
| 0.1 | -3.6728e-002 | 0. | -3.3707e-003 |
| 0.175 | -0.10882 | | -1.0586e-002 |
| 0.2875 | -0.24548 | | -3.1683e-002 |
| 0.45625 | -0.46599 | | -7.3882e-002 |
| 0.65625 | -0.72352 | | -0.13264 |
| 0.85625 | -0.97979 | | -0.199 |
| 1. | -1.1627 | | -0.25089 |

**FIGURE 9**
**Model (A4) > Static Structural (A5) > Solution (A6) > Directional Deformation 3 > Figure**

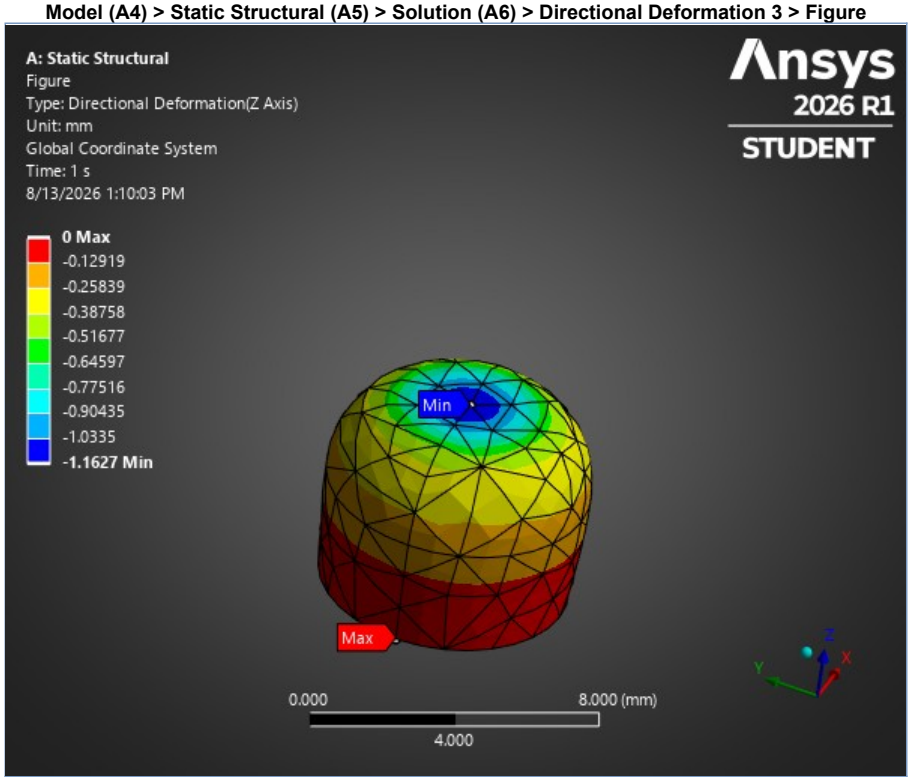



**FIGURE 10**
**Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Elastic Strain**

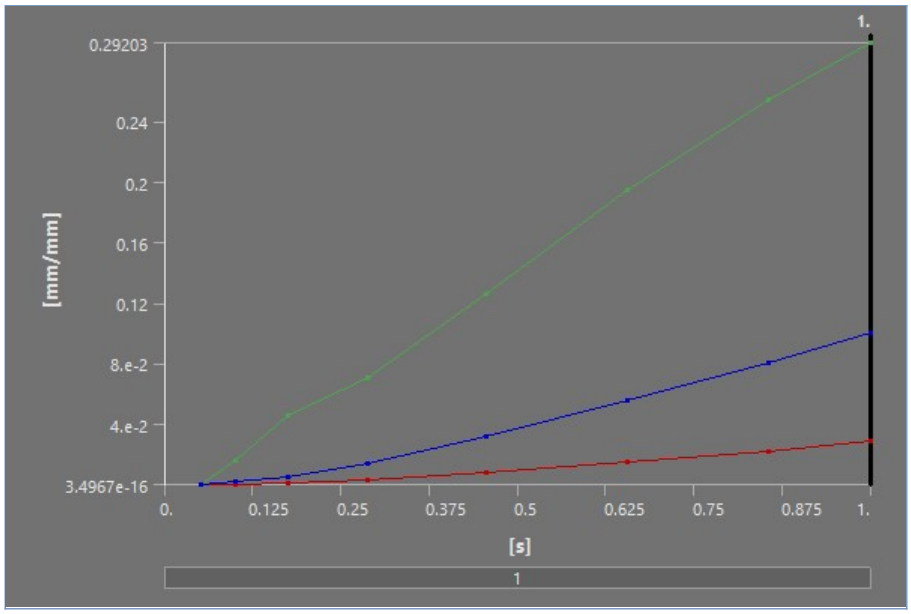



**TABLE 23**
**Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Elastic Strain**

| Time [s] | Minimum [mm/mm] | Maximum [mm/mm] | Average [mm/mm] |
|---|---|---|---|
| 5.e-002 | 3.4967e-016 | 3.7822e-015 | 1.4812e-015 |
| 0.1 | 3.2282e-004 | 1.627e-002 | 1.6481e-003 |
| 0.175 | 1.0321e-003 | 4.5708e-002 | 5.1146e-003 |
| 0.2875 | 3.2183e-003 | 7.0992e-002 | 1.4385e-002 |
| 0.45625 | 7.7978e-003 | 0.1262 | 3.206e-002 |
| 0.65625 | 1.4432e-002 | 0.19444 | 5.5409e-002 |
| 0.85625 | 2.2297e-002 | 0.25443 | 8.0825e-002 |
| 1. | 2.8821e-002 | 0.29203 | 0.10035 |

**FIGURE 11**
**Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Elastic Strain > Figure**

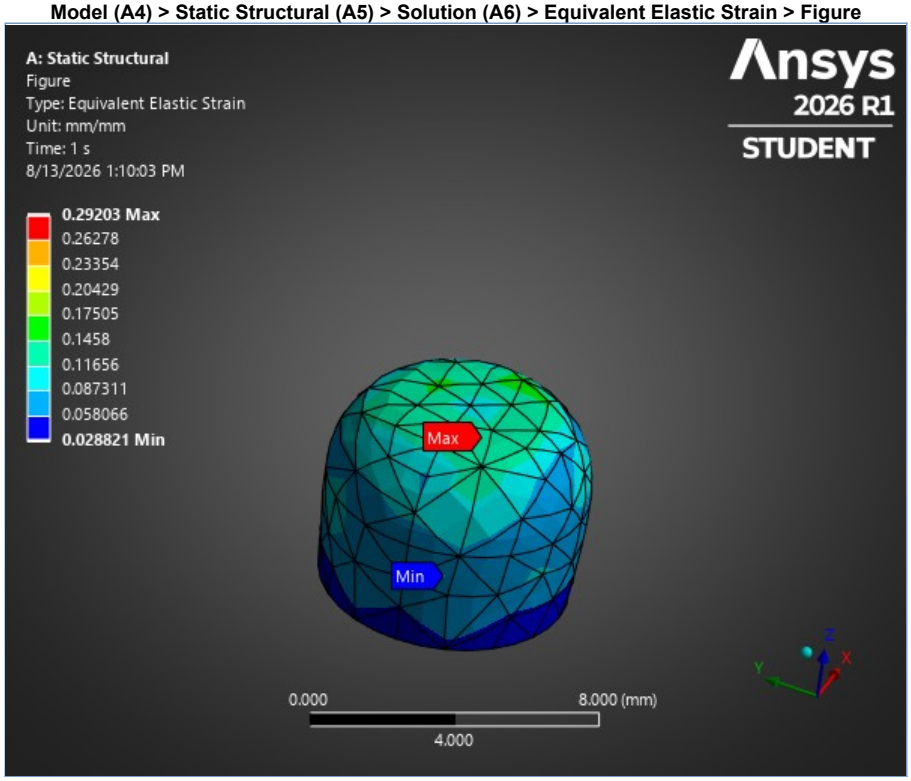



**FIGURE 12**
**Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Principal Elastic Strain**

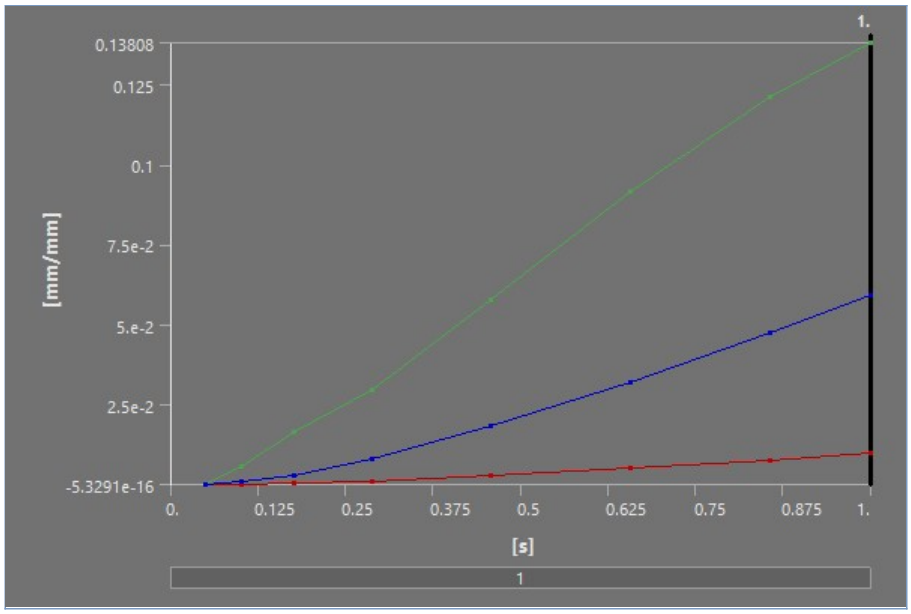



**TABLE 24**
**Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Principal Elastic Strain**

| Time [s] | Minimum [mm/mm] | Maximum [mm/mm] | Average [mm/mm] |
|---|---|---|---|
| 5.e-002 | -5.3291e-016 | 2.5535e-015 | 3.6837e-016 |
| 0.1 | 1.1258e-004 | 5.8647e-003 | 9.0284e-004 |
| 0.175 | 3.5857e-004 | 1.6331e-002 | 2.8157e-003 |
| 0.2875 | 1.1143e-003 | 2.9491e-002 | 8.0922e-003 |
| 0.45625 | 2.6952e-003 | 5.7715e-002 | 1.8373e-002 |
| 0.65625 | 4.994e-003 | 9.1593e-002 | 3.2147e-002 |
| 0.85625 | 7.7278e-003 | 0.12118 | 4.7308e-002 |
| 1. | 9.98e-003 | 0.13808 | 5.9054e-002 |

**FIGURE 13**
**Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Principal Elastic Strain > Figure**

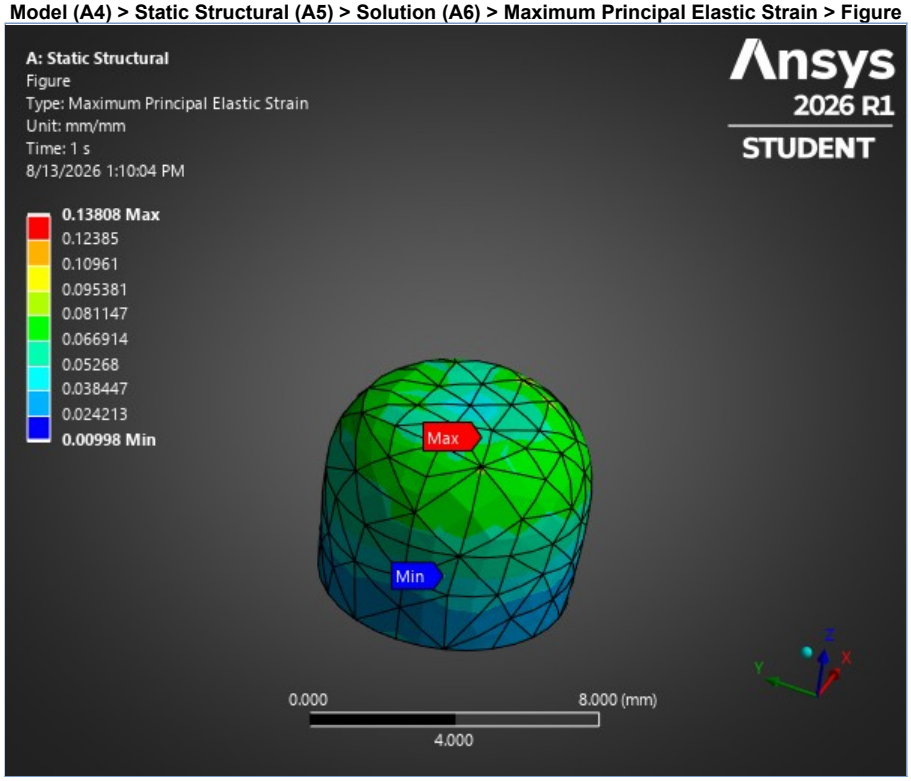



**FIGURE 14**
**Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Shear Elastic Strain**

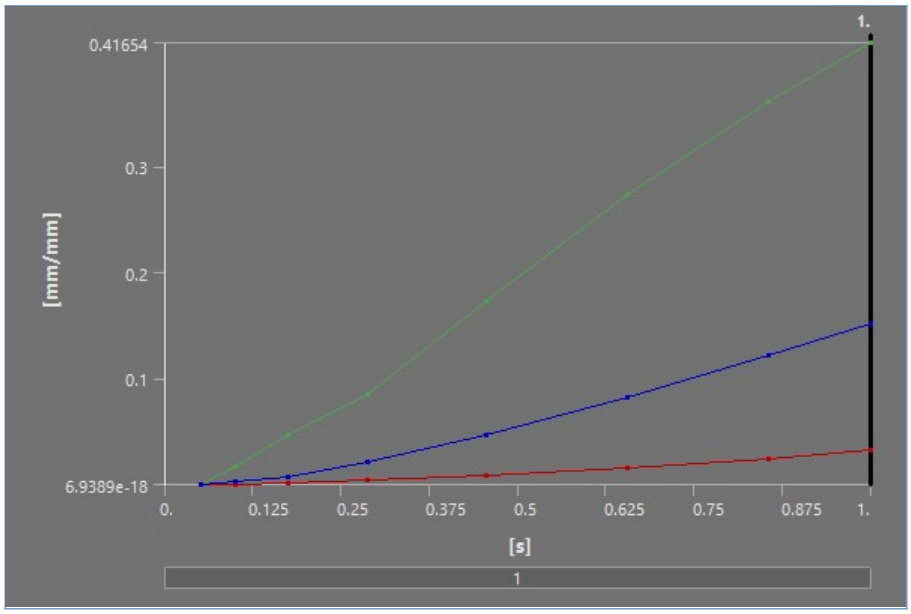



**TABLE 25**
**Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Shear Elastic Strain**

| Time [s] | Minimum [mm/mm] | Maximum [mm/mm] | Average [mm/mm] |
|---|---|---|---|
| 5.e-002 | 6.9389e-018 | 5.033e-015 | 8.5159e-016 |
| 0.1 | 3.5933e-004 | 1.6886e-002 | 2.3153e-003 |
| 0.175 | 1.1456e-003 | 4.723e-002 | 7.233e-003 |
| 0.2875 | 3.5614e-003 | 8.5388e-002 | 2.0919e-002 |
| 0.45625 | 8.6228e-003 | 0.17325 | 4.7423e-002 |
| 0.65625 | 1.5991e-002 | 0.27318 | 8.2741e-002 |
| 0.85625 | 2.4775e-002 | 0.3619 | 0.12151 |
| 1. | 3.2066e-002 | 0.41654 | 0.15148 |

**FIGURE 15**
**Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Shear Elastic Strain > Figure**

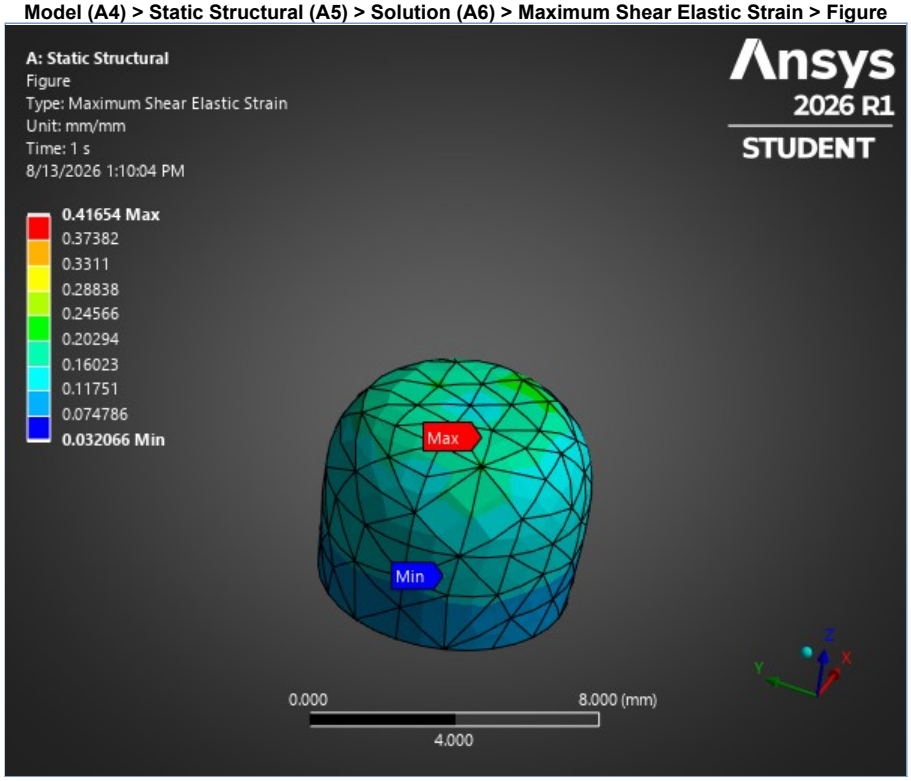



**FIGURE 16**
**Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress**

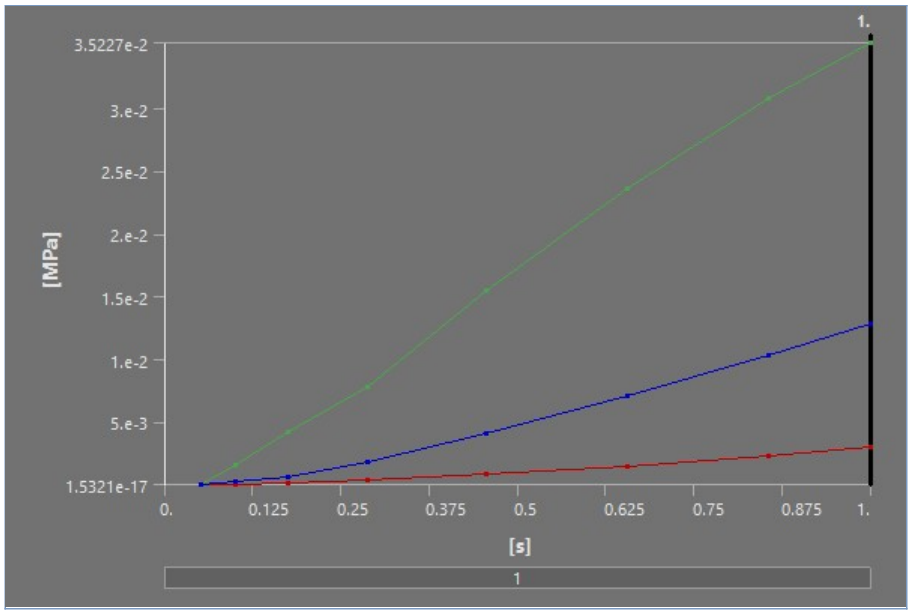



**TABLE 26**
**Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress**

| Time [s] | Minimum [MPa] | Maximum [MPa] | Average [MPa] |
|---|---|---|---|
| 5.e-002 | 1.5321e-017 | 6.1801e-016 | 1.1367e-016 |
| 0.1 | 3.3847e-005 | 1.5262e-003 | 2.037e-004 |
| 0.175 | 1.079e-004 | 4.2153e-003 | 6.3482e-004 |
| 0.2875 | 3.3508e-004 | 7.7732e-003 | 1.8315e-003 |
| 0.45625 | 8.0973e-004 | 1.5422e-002 | 4.1163e-003 |
| 0.65625 | 1.4968e-003 | 2.3664e-002 | 7.1112e-003 |
| 0.85625 | 2.3099e-003 | 3.0784e-002 | 1.0346e-002 |
| 1. | 2.982e-003 | 3.5227e-002 | 1.2817e-002 |

**FIGURE 17**
**Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress > Figure**

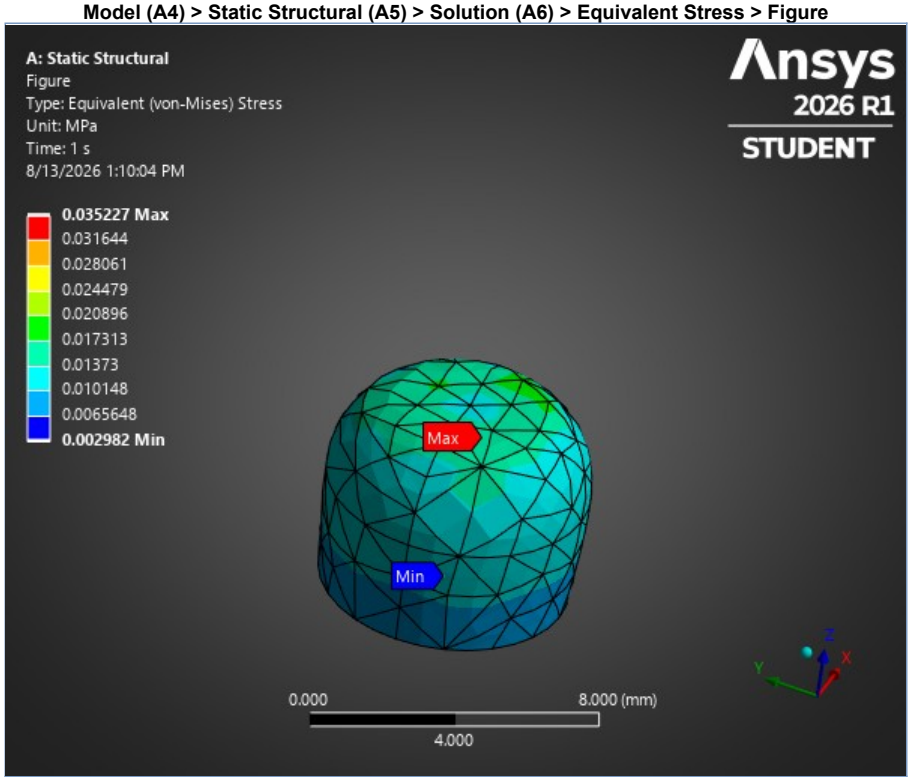



**FIGURE 18**
**Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Principal Stress**

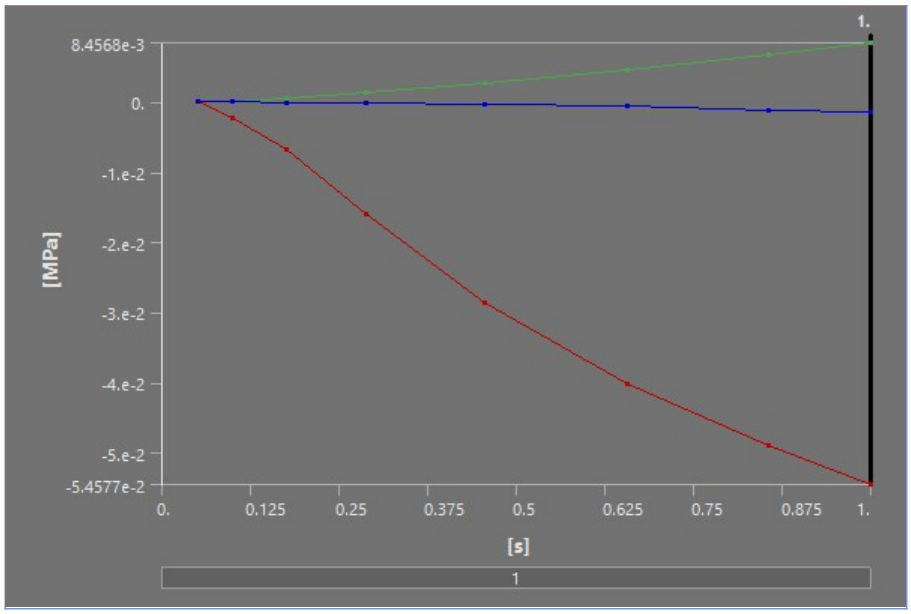



**TABLE 27**
**Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Principal Stress**

| Time [s] | Minimum [MPa] | Maximum [MPa] | Average [MPa] |
|---|---|---|---|
| 5.e-002 | -9.3872e-015 | 7.3559e-015 | -1.8816e-016 |
| 0.1 | -2.3292e-003 | 1.9808e-004 | -7.3429e-006 |
| 0.175 | -6.8524e-003 | 5.9549e-004 | -2.5307e-005 |
| 0.2875 | -1.5952e-002 | 1.3809e-003 | -1.1316e-004 |
| 0.45625 | -2.8738e-002 | 2.6118e-003 | -3.0571e-004 |
| 0.65625 | -4.0208e-002 | 4.5248e-003 | -6.4306e-004 |
| 0.85625 | -4.9075e-002 | 6.7701e-003 | -1.0941e-003 |
| 1. | -5.4577e-002 | 8.4568e-003 | -1.482e-003 |

**FIGURE 19**
**Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Principal Stress > Figure**

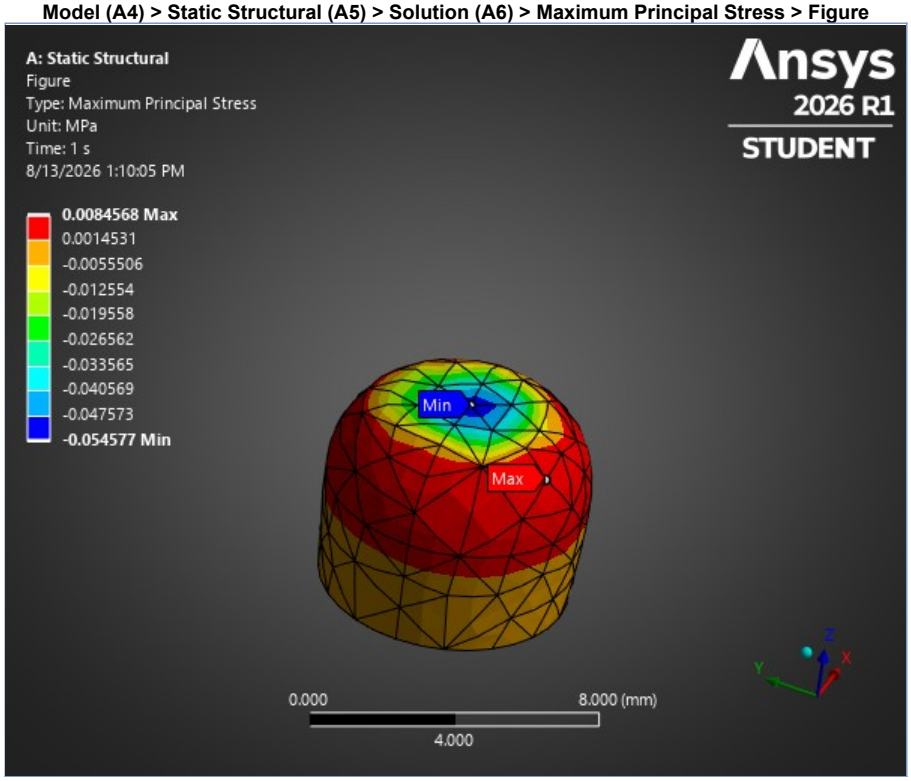



**FIGURE 20**
**Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Shear Stress**

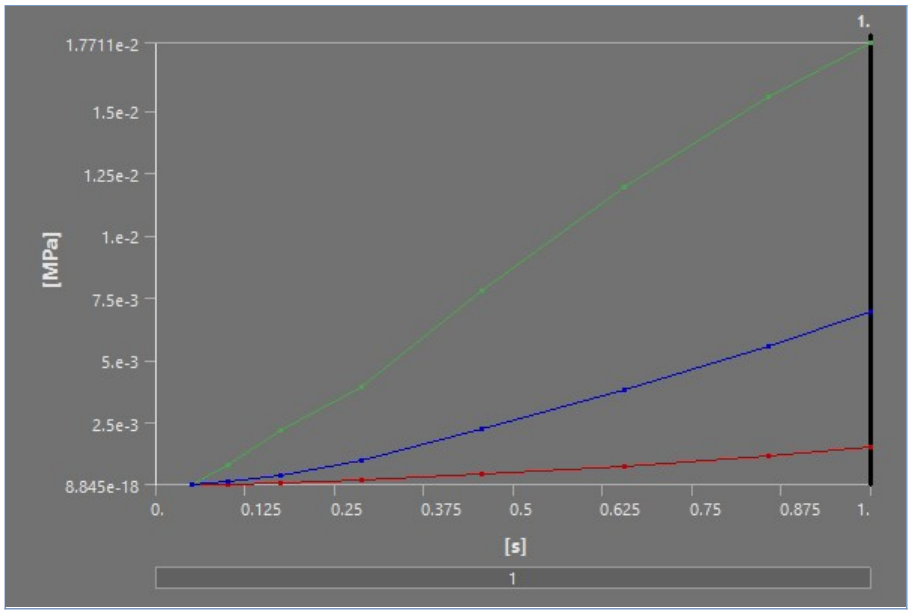



**TABLE 28**
**Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Shear Stress**

| Time [s] | Minimum [MPa] | Maximum [MPa] | Average [MPa] |
|---|---|---|---|
| 5.e-002 | 8.845e-018 | 3.4715e-016 | 6.3919e-017 |
| 0.1 | 1.6993e-005 | 7.9309e-004 | 1.0934e-004 |
| 0.175 | 5.4156e-005 | 2.1915e-003 | 3.4062e-004 |
| 0.2875 | 1.6817e-004 | 3.917e-003 | 9.8043e-004 |
| 0.45625 | 4.062e-004 | 7.7521e-003 | 2.2065e-003 |
| 0.65625 | 7.5076e-004 | 1.1958e-002 | 3.8192e-003 |
| 0.85625 | 1.1585e-003 | 1.5568e-002 | 5.5674e-003 |
| 1. | 1.4946e-003 | 1.7711e-002 | 6.9066e-003 |

**FIGURE 21**
**Model (A4) > Static Structural (A5) > Solution (A6) > Maximum Shear Stress > Figure**

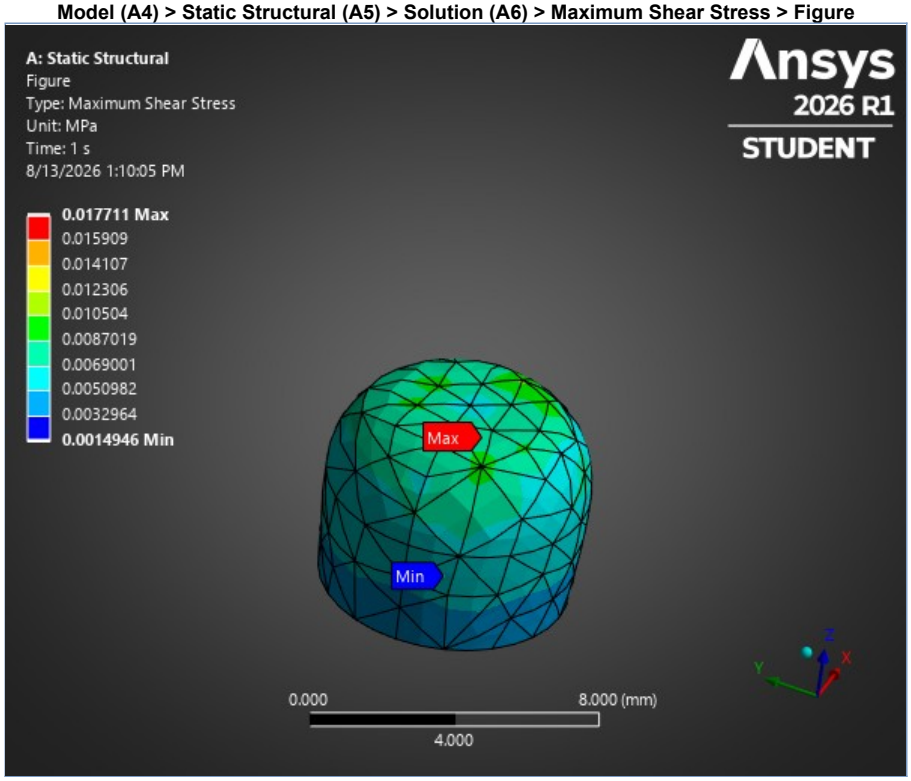



**FIGURE 22**
**Model (A4) > Static Structural (A5) > Solution (A6) > Stress Intensity**

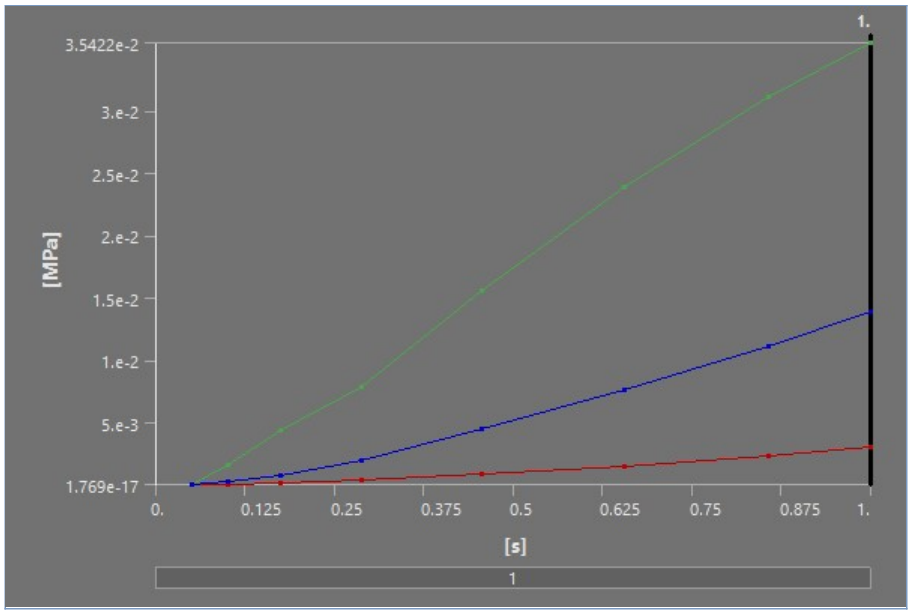



**TABLE 29**
**Model (A4) > Static Structural (A5) > Solution (A6) > Stress Intensity**

| Time [s] | Minimum [MPa] | Maximum [MPa] | Average [MPa] |
|---|---|---|---|
| 5.e-002 | 1.769e-017 | 6.943e-016 | 1.2784e-016 |
| 0.1 | 3.3987e-005 | 1.5862e-003 | 2.1869e-004 |
| 0.175 | 1.0831e-004 | 4.383e-003 | 6.8124e-004 |
| 0.2875 | 3.3634e-004 | 7.834e-003 | 1.9609e-003 |
| 0.45625 | 8.1241e-004 | 1.5504e-002 | 4.413e-003 |
| 0.65625 | 1.5015e-003 | 2.3916e-002 | 7.6384e-003 |
| 0.85625 | 2.317e-003 | 3.1136e-002 | 1.1135e-002 |
| 1. | 2.9891e-003 | 3.5422e-002 | 1.3813e-002 |

**FIGURE 23**
**Model (A4) > Static Structural (A5) > Solution (A6) > Stress Intensity > Figure**

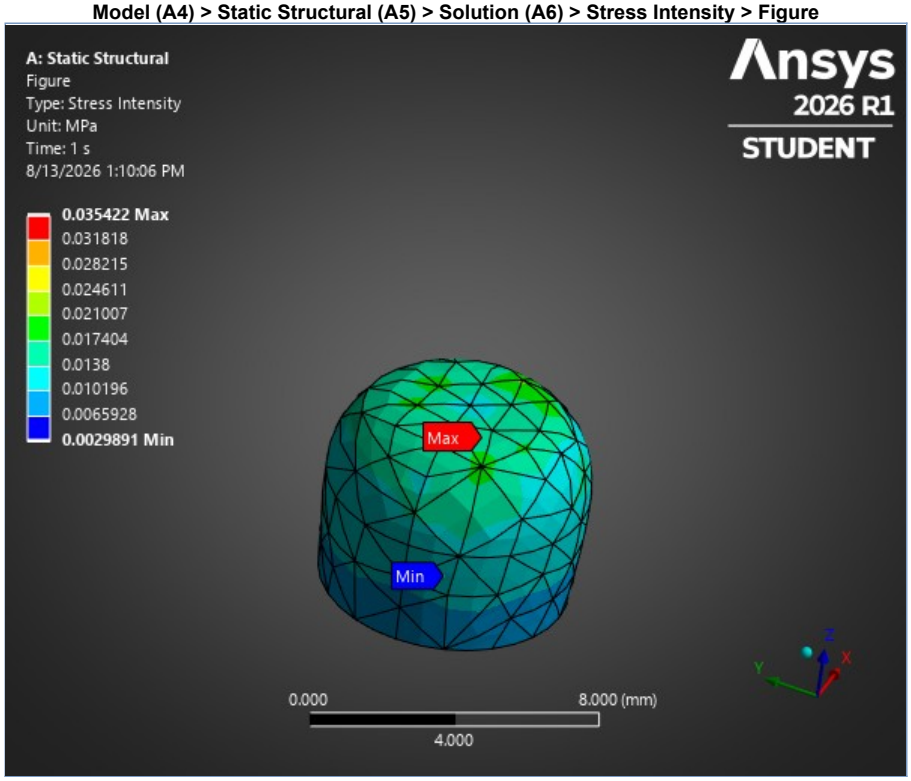



**TABLE 30**
**Model (A4) > Static Structural (A5) > Solution (A6) > Results**

| Object Name | *Strain Energy* |
|---|---|
| State | Solved |
| **Scope** | |
| Scoping Method | Geometry Selection |
| Geometry | 1 Body |
| **Definition** | |
| Type | Strain Energy |
| By | Time |
| Display Time | Last |
| Separate Data by Entity | No |
| Calculate Time History | Yes |
| Identifier | |
| Suppressed | No |
| **Results** | |
| Minimum | 5.7516e-005 mJ |
| Maximum | 7.9832e-003 mJ |
| Total | 0.25645 mJ |
| Minimum Occurs On | run 6 solid\|Body3 |
| Maximum Occurs On | run 6 solid\|Body3 |
| **Minimum Value Over Time** | |
| Minimum | -2.3216e-017 mJ |
| Maximum | 5.7516e-005 mJ |
| **Maximum Value Over Time** | |
| Minimum | 1.0388e-017 mJ |
| Maximum | 7.9832e-003 mJ |
| **Information** | |
| Time | 1. s |
| Load Step | 1 |
| Substep | 8 |
| Iteration Number | 29 |

**FIGURE 24**
**Model (A4) > Static Structural (A5) > Solution (A6) > Strain Energy**

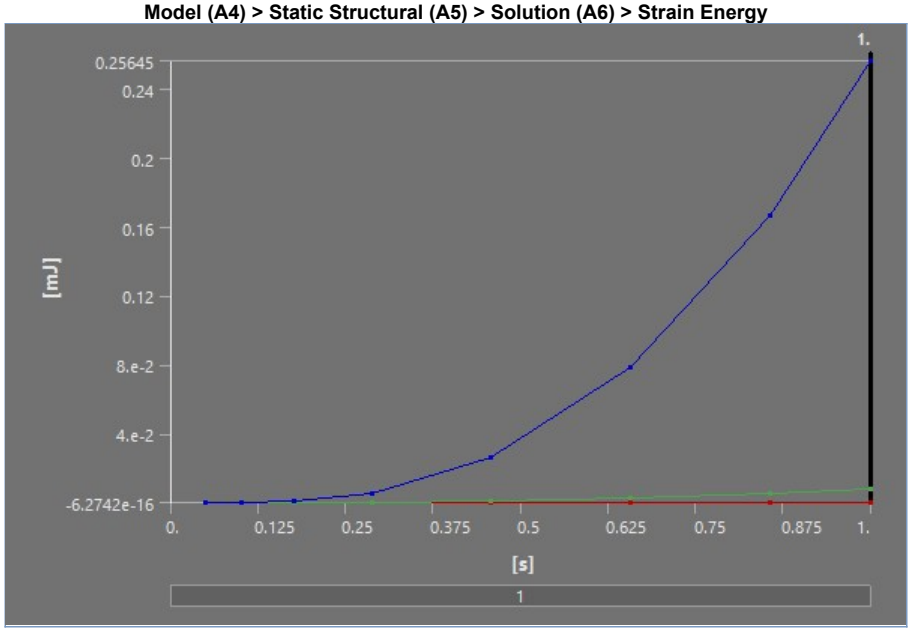



**TABLE 31**
**Model (A4) > Static Structural (A5) > Solution (A6) > Strain Energy**

| Time [s] | Minimum [mJ] | Maximum [mJ] | Total [mJ] |
|---|---|---|---|
| 5.e-002 | -2.3216e-017 | 1.0388e-017 | -6.2742e-016 |
| 0.1 | 1.0784e-008 | 5.5402e-006 | 8.3498e-005 |
| 0.175 | 9.6414e-008 | 4.6311e-005 | 7.7952e-004 |
| 0.2875 | 7.5892e-007 | 1.7521e-004 | 5.5782e-003 |
| 0.45625 | 4.115e-006 | 7.8675e-004 | 2.658e-002 |
| 0.65625 | 1.3937e-005 | 2.4358e-003 | 7.8589e-002 |
| 0.85625 | 3.3726e-005 | 5.2245e-003 | 0.16684 |
| 1. | 5.7516e-005 | 7.9832e-003 | 0.25645 |

**FIGURE 25**

**Model (A4) > Static Structural (A5) > Solution (A6) > Strain Energy > Figure**

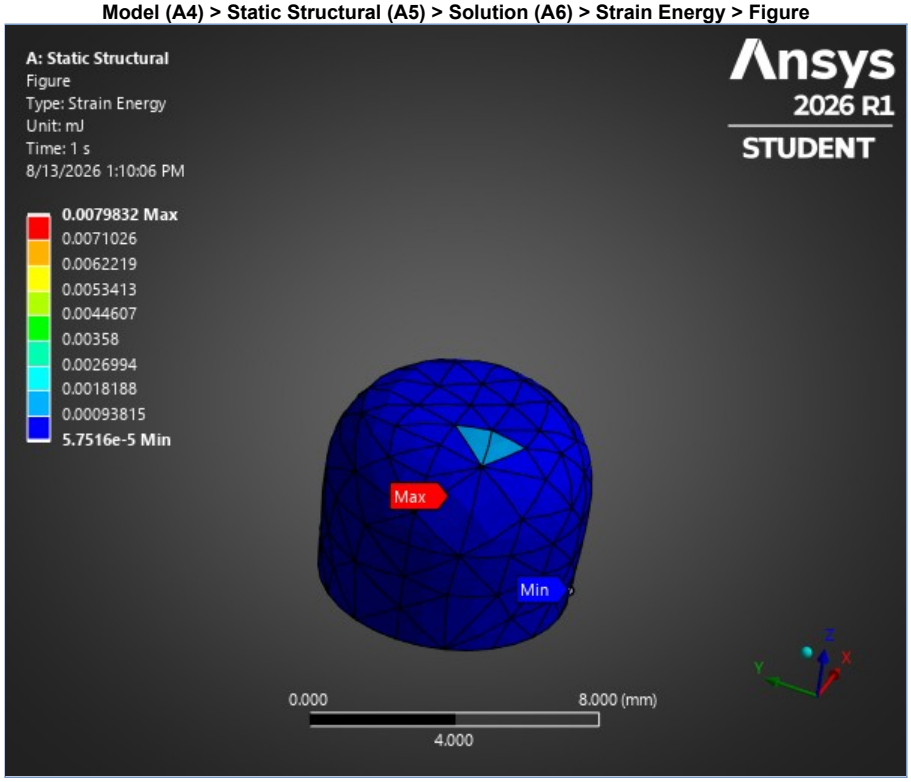



**TABLE 32**
**Model (A4) > Static Structural (A5) > Solution (A6) > Contact Tools**

| Object Name | *Contact Tool* |
|---|---|
| State | Solved |
| **Scope** | |
| Scoping Method | Worksheet |

**Model (A4) > Static Structural (A5) > Solution (A6) > Contact Tool**

| Name | Contact Side |
|---|---|
| Contact Region | Both |
| Frictionless - run 6 solid\|Body3 To run 6 solid\|Body2 | Both |

**TABLE 33**
**Model (A4) > Static Structural (A5) > Solution (A6) > Contact Tool > Results**

| Object Name | *Status* |
|---|---|
| State | Solved |
| **Definition** | |
| Type | Status |
| By | Time |
| Display Time | Last |
| Calculate Time History | Yes |
| Identifier | |
| Suppressed | No |
| **Integration Point Results** | |
| Display Option | Averaged |
| **Information** | |
| Time | 1. s |
| Load Step | 1 |
| Substep | 8 |
| Iteration Number | 29 |

**FIGURE 26**
**Model (A4) > Static Structural (A5) > Solution (A6) > Contact Tool > Status**

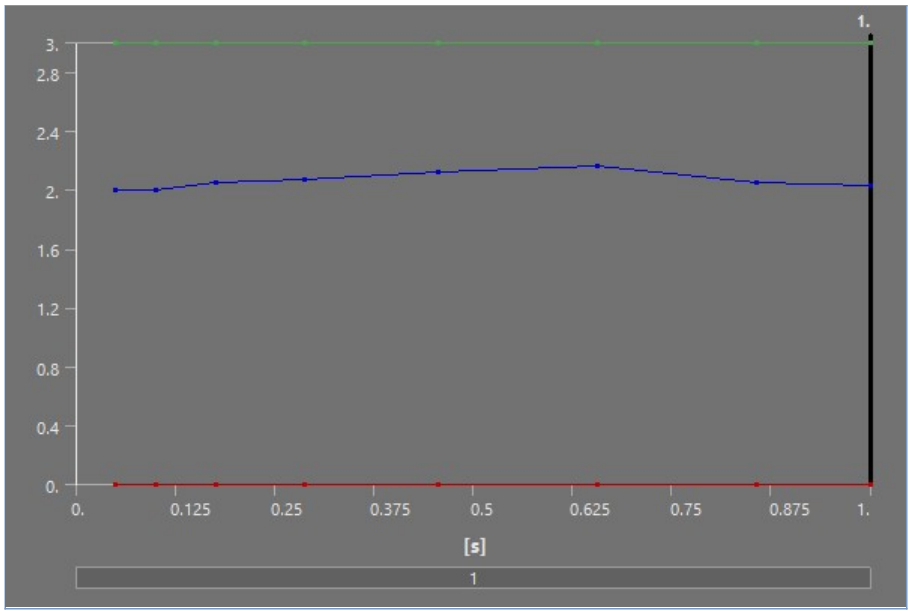



**TABLE 34**
**Model (A4) > Static Structural (A5) > Solution (A6) > Contact Tool > Status**

| Time [s] | Minimum | Maximum | Average |
|---|---|---|---|
| 5.e-002 | 0. | 3. | 2.002 |
| 0.1 | | | 2.0019 |
| 0.175 | | | 2.0463 |
| 0.2875 | | | 2.071 |
| 0.45625 | | | 2.1228 |
| 0.65625 | | | 2.167 |
| 0.85625 | | | 2.0528 |
| 1. | | | 2.0295 |

**FIGURE 27**
**Model (A4) > Static Structural (A5) > Solution (A6) > Contact Tool > Figure**

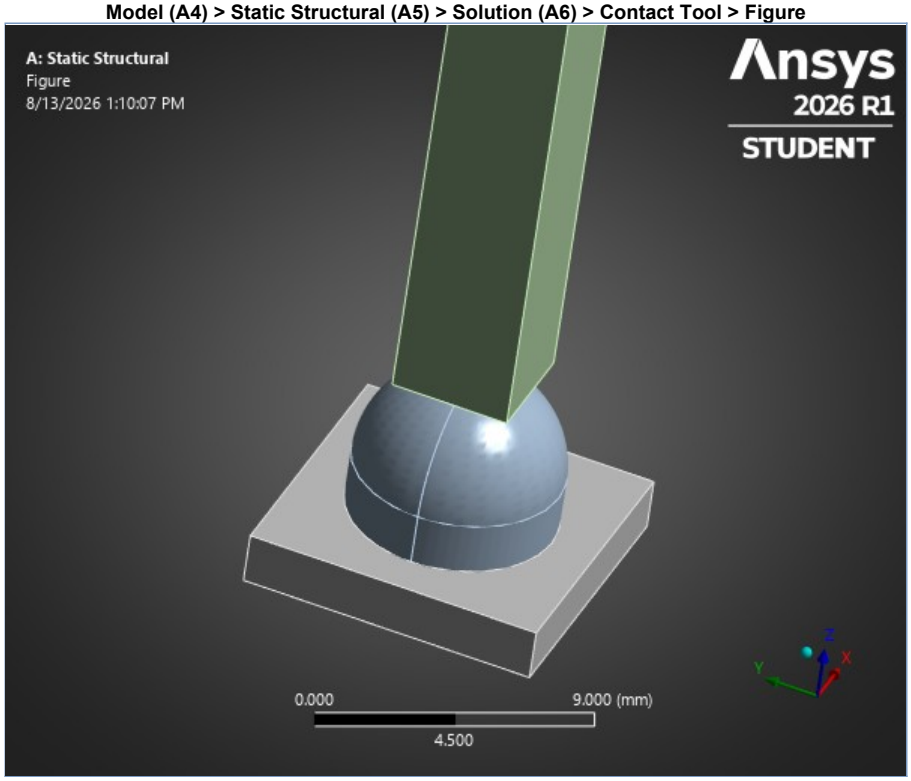



**TABLE 35**
**Model (A4) > Static Structural (A5) > Solution (A6) > Probes**

| Object Name | *Force Reaction* | *Moment Reaction* |
|---|---|---|
| State | Solved | |
| **Definition** | | |
| Type | Force Reaction | Moment Reaction |
| Location Method | Boundary Condition | |
| Boundary Condition | Displacement | |
| Orientation | Global Coordinate System | |
| Suppressed | No | |
| Summation | | Centroid |
| **Options** | | |
| Result Selection | All | |
| Display Time | End Time | |
| **Results** | | |
| X Axis | -1.9946e-003 N | -1.1061e-002 N·mm |
| Y Axis | -2.6381e-003 N | 2.278e-003 N·mm |
| Z Axis | -0.61325 N | 1.5399e-004 N·mm |
| Total | 0.61326 N | 1.1294e-002 N·mm |
| **Maximum Value Over Time** | | |
| X Axis | 1.1104e-003 N | 9.9896e-005 N·mm |
| Y Axis | 1.0224e-003 N | 2.278e-003 N·mm |
| Z Axis | -2.4513e-006 N | 1.5399e-004 N·mm |
| Total | 0.61326 N | 1.1294e-002 N·mm |
| **Minimum Value Over Time** | | |
| X Axis | -1.9946e-003 N | -1.1061e-002 N·mm |
| Y Axis | -2.6381e-003 N | -6.8762e-003 N·mm |
| Z Axis | -0.61325 N | -6.3835e-005 N·mm |
| Total | 2.452e-006 N | 1.066e-006 N·mm |
| **Information** | | |
| Time | 1. s | |
| Load Step | 1 | |
| Substep | 8 | |
| Iteration Number | 29 | |

**FIGURE 28**
**Model (A4) > Static Structural (A5) > Solution (A6) > Force Reaction**

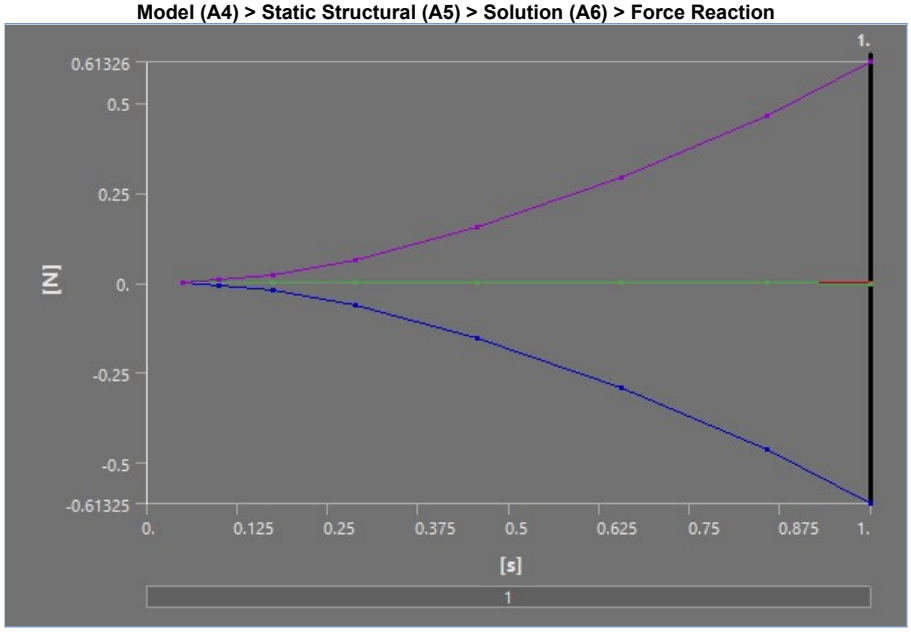



**TABLE 36**
**Model (A4) > Static Structural (A5) > Solution (A6) > Force Reaction**

| Time [s] | Force Reaction (X) [N] | Force Reaction (Y) [N] | Force Reaction (Z) [N] | Force Reaction (Total) [N] |
|---|---|---|---|---|
| 5.e-002 | -3.9804e-008 | -3.9805e-008 | -2.4513e-006 | 2.452e-006 |
| 0.1 | 3.3077e-005 | -3.4653e-005 | -6.3665e-003 | 6.3667e-003 |
| 0.175 | -4.4591e-006 | -5.2745e-005 | -2.0426e-002 | 2.0426e-002 |
| 0.2875 | -1.1153e-004 | 1.6981e-004 | -6.396e-002 | 6.3961e-002 |
| 0.45625 | -7.2806e-004 | 2.5801e-004 | -0.15547 | 0.15547 |
| 0.65625 | 1.1104e-003 | 1.0224e-003 | -0.2932 | 0.2932 |
| 0.85625 | -7.8647e-004 | -1.0958e-003 | -0.46501 | 0.46501 |
| 1. | -1.9946e-003 | -2.6381e-003 | -0.61325 | 0.61326 |

**FIGURE 29**
**Model (A4) > Static Structural (A5) > Solution (A6) > Force Reaction > Figure**

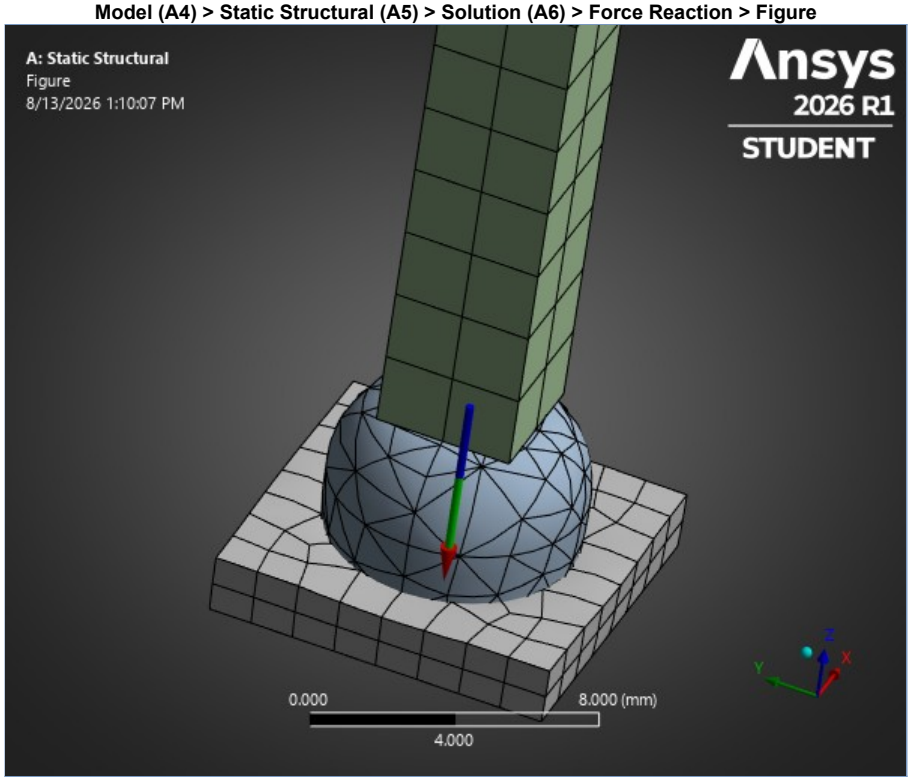



**FIGURE 30**
**Model (A4) > Static Structural (A5) > Solution (A6) > Moment Reaction**

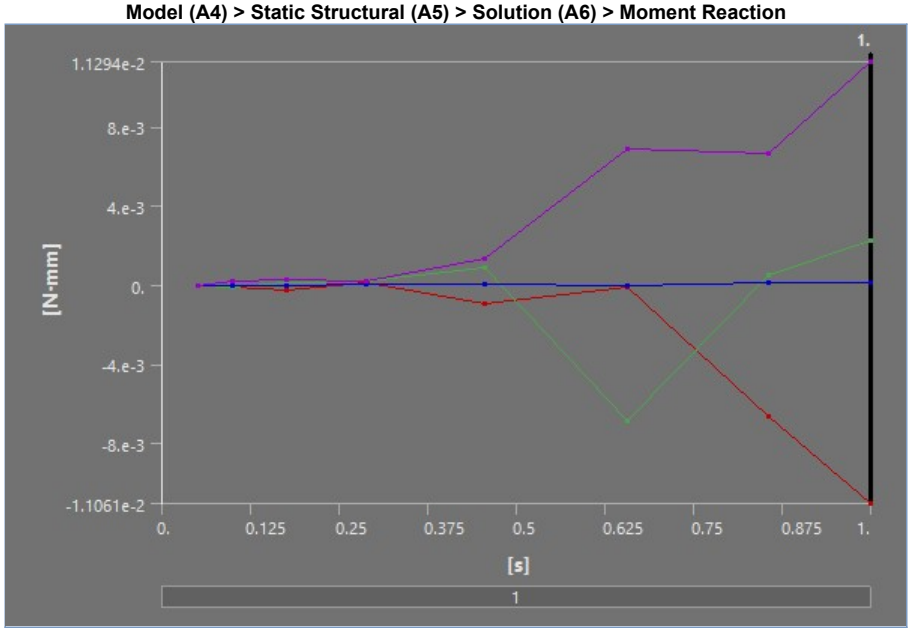
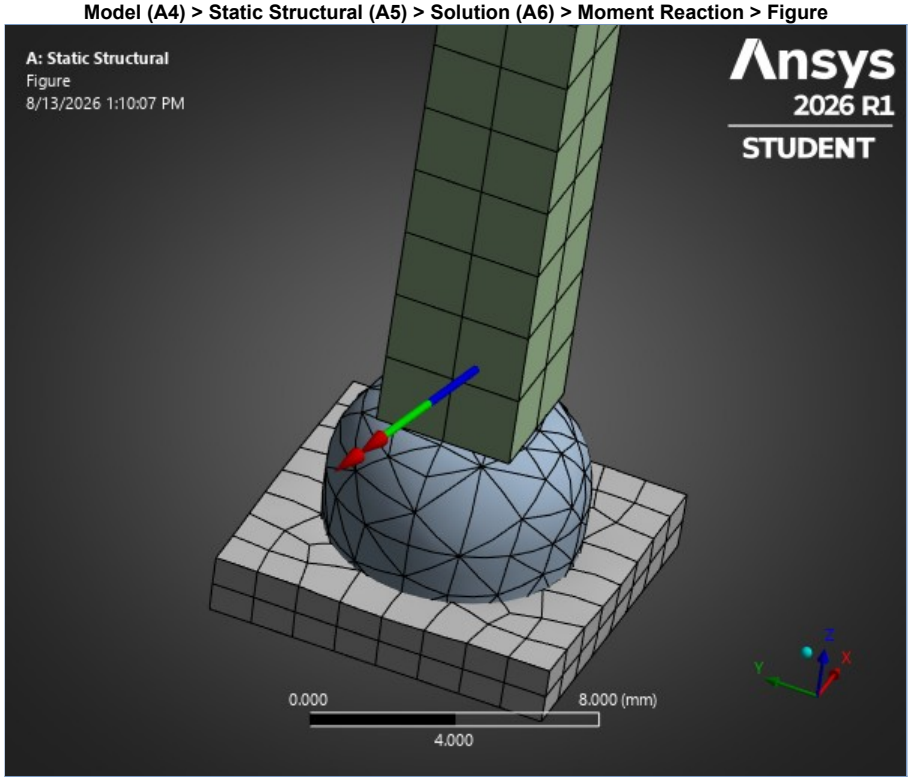



**TABLE 37**
**Model (A4) > Static Structural (A5) > Solution (A6) > Moment Reaction**

| Time [s] | Moment Reaction (X) [N·mm] | Moment Reaction (Y) [N·mm] | Moment Reaction (Z) [N·mm] | Moment Reaction (Total) [N·mm] |
|---|---|---|---|---|
| 5.e-002 | 7.538e-007 | -7.538e-007 | 7.8348e-014 | 1.066e-006 |
| 0.1 | -1.2448e-004 | -9.9482e-005 | 1.7514e-007 | 1.5935e-004 |
| 0.175 | -2.411e-004 | 1.1675e-004 | -2.6985e-006 | 2.6789e-004 |
| 0.2875 | 9.9896e-005 | 1.873e-004 | 3.2631e-006 | 2.123e-004 |
| 0.45625 | -9.684e-004 | 8.6439e-004 | 2.3784e-005 | 1.2983e-003 |
| 0.65625 | -9.9917e-005 | -6.8762e-003 | -6.3835e-005 | 6.8772e-003 |
| 0.85625 | -6.6495e-003 | 5.1671e-004 | 1.1035e-004 | 6.6704e-003 |
| 1. | -1.1061e-002 | 2.278e-003 | 1.5399e-004 | 1.1294e-002 |

**FIGURE 31**
**Model (A4) > Static Structural (A5) > Solution (A6) > Moment Reaction > Figure**




# Material Data

## *ecoflex 00-45*

**TABLE 38**
**ecoflex 00-45 > Constants**

| | |
|---|---|
| Density | 1.06e-006 kg mm^-3 |

**TABLE 39**
**ecoflex 00-45 > Color**

| Red | Green | Blue |
|---|---|---|
| 109 | 157 | 209 |

**TABLE 40**
**ecoflex 00-45 > Neo-Hookean**

| Initial Shear Modulus Mu MPa | Incompressibility Parameter D1 MPa^-1 | Temperature K |
|---|---|---|
| 4.73e-002 | 0.424 | |

## *Plastic, ABS (high-impact)*

**TABLE 41**
**Plastic, ABS (high-impact) > Constants**

| | |
|---|---|
| Density | 1.03e-006 kg mm^-3 |
| Tensile Yield Strength | 43.82 MPa |
| Tensile Ultimate Strength | 36.26 MPa |
| Coefficient of Thermal Expansion | 1.84e-004 K^-1 |
| Thermal Conductivity | 1.997e-004 W mm^-1 K^-1 |
| Specific Heat | 1.4e+006 mJ kg^-1 K^-1 |
| Resistivity | 9.95e+016 ohm mm |
| Electric Loss Tangent | 5.916e-003 |
| Relative Permittivity | 2.993 |

**TABLE 42**
**Plastic, ABS (high-impact) > Opacity**

| Red | Green | Blue |
|---|---|---|
| 0 | 153 | 255 |

| Opacity |
|---|
| 0.9 |

| Metallic Finish |
|---|
| 0 |

**TABLE 43**
**Plastic, ABS (high-impact) > Isotropic Elasticity**

| Young's Modulus MPa | Poisson's Ratio | Bulk Modulus MPa | Shear Modulus MPa | Temperature K |
|---|---|---|---|---|
| 2405 | 0.4089 | 4399.9 | 853.5 | 296.15 |

**TABLE 44**
**Plastic, ABS (high-impact) > Isotropic Secant Coefficient of Thermal Expansion**

| Zero-Thermal-Strain Reference Temperature K |
|---|
| 296.15 |

**TABLE 45**
**Plastic, ABS (high-impact) > Climate change (CO2-eq)**

| Climate change (CO2-eq) dBA |
|---|
| 3.661 |

**TABLE 46**
**Plastic, ABS (high-impact) > Embodied energy**

| Embodied energy mJ kg^-1 |
|---|
| 9.64e+010 |

**TABLE 47**
**Plastic, ABS (high-impact) > Recycle**

| Recycle |
|---|
| 1 |